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1. Discrete Random Variables

$$P(X=x) = \begin{cases} k(2x+1), & x = 1, 2, 3, 4 \\ 0, & \text{otherwise} \end{cases}$$

(a) Find the constant value of k

* The probabilities of a PMF must add up to 1

$$P(X=1) = k(2(1)+1) = 3k$$

$$P(X=2) = k(2(2)+1) = 5k$$

$$P(X=3) = k(2(3)+1) = 7k$$

$$P(X=4) = k(2(4)+1) = 9k$$

$$3k + 5k + 7k + 9k = 1$$

$$k = \frac{1}{24}$$

$$\text{check: } \frac{3}{24} + \frac{5}{24} + \frac{7}{24} + \frac{9}{24} = \frac{24}{24} = 1$$

(b) compute the expected value $E[X]$ and variance $\text{Var}(X)$

Expected value is a weighted average:

$$E[X] = \sum x \cdot P(X=x)$$

$$= (1) \left(\frac{3}{24}\right) + (2) \left(\frac{5}{24}\right) + (3) \left(\frac{7}{24}\right) + (4) \left(\frac{9}{24}\right)$$

$$E[X] = \frac{35}{12} \approx 2.9167$$

$$\text{Var}(X) = E(X^2) - (E[X])^2$$

$$E(X^2) = \sum x^2 P(X=x)$$

$$= (1)^2 \left(\frac{3}{24}\right) + (2)^2 \left(\frac{5}{24}\right) + (3)^2 \left(\frac{7}{24}\right) + (4)^2 \left(\frac{9}{24}\right)$$

$$= \frac{115}{12} \approx 9.5833$$

$$(E[X])^2 = \left(\frac{35}{12}\right)^2 = 8.5069$$

$$\text{Var}(X) = 9.5833 - 8.5069$$

$$\boxed{\text{Var}(X) = 1.0764}$$

(c) Calculate $P(X \geq 3)$

$$P(X \geq 3) = P(X=3) + P(X=4)$$

$$= \frac{7}{24} + \frac{9}{24}$$

$$\boxed{P(X \geq 3) = \frac{2}{3} \approx 0.6667}$$

2. Binomial Distribution

(a) What is the probability that exactly 2 requests are malicious?

Given: $n = 20$ requests

$$p(\text{malicious}) = 0.08$$

Let X = number of malicious requests

$$P(X=x) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$P(X=2) = \binom{20}{2} (0.08)^2 (1-0.08)^{20-2}$$

$$* \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

$$= \frac{20!}{2!(20-2)!} = 190$$

$$P(X=2) = 190 (0.08)^2 (0.92)^{18}$$

$$P(X=2) = 0.2711 \approx 27.11\%$$

(b) What is the probability that at least 3 requests are malicious?

$$P(X \geq 3) = 1 - P(X < 3)$$

$$= 1 - [P(X=0) + P(X=1) + P(X=2)]$$

$$* P(X=0) = \binom{20}{0} (0.08)^0 (0.92)^{20} = 0.1887$$

$$* P(X=1) = \binom{20}{1} (0.08)^1 (0.92)^{20-1} = 0.3282$$

$$* P(X=2) = \binom{20}{2} (0.08)^2 (0.92)^{20-2} = 0.2711$$

$$= 1 - (0.1887 + 0.3282 + 0.2711)$$

$$P(X \geq 3) = 0.212 \approx 21.2\%$$

(c) Determine the mean and standard deviation of the number of malicious requests.

$$\text{mean } \mu = np \quad \text{SD } \sigma = \sqrt{np(1-p)}$$

$$= (20)(0.08) \quad = \sqrt{20(0.08)(0.92)}$$

$$\mu = 1.6$$

$$\sigma = 1.2133$$

3. Poisson Distribution

(a) Find the probability of receiving exactly 6 hits in a given minute.

Given: $\lambda = 4.5$

• Poisson Distribution Formula:

$$P(X=r) = \frac{e^{-\lambda} \lambda^r}{r!}$$

where $P(X=r) \rightarrow$ probability of observing r events

$\lambda \rightarrow$ average rate of occurrence

$r \rightarrow$ no. of events that occur

$$P(X=6) = \frac{e^{-4.5} (4.5)^6}{6!} = 0.1281 \text{ or } 12.81\%$$

(b) Find the probability of receiving at most 2 hits in a given minute.

$$P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$$

$$* P(X=0) = \frac{e^{-4.5} (4.5)^0}{0!} = 0.0111$$

$$* P(X=1) = \frac{e^{-4.5} (4.5)^1}{1!} = 0.0500$$

$$* P(X=2) = \frac{e^{-4.5} (4.5)^2}{2!} = 0.1125$$

$$P(X \leq 2) = 0.0111 + 0.0500 + 0.1125 = 0.1736 \text{ or } 17.36\%$$

(c) Calculate the probability of receiving more than 10 hits in a 2-minute interval

$$P(X > 10) = 1 - P(X \leq 10)$$

$$= 1 - [P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) + P(X=5) + P(X=6) + P(X=7) + P(X=8) + P(X=9) + P(X=10)]$$

$$P(X=1) = \frac{e^{-9} (9)^1}{1!} = 0.0001$$

$$P(X=2) = \frac{e^{-9} (9)^2}{2!} = 0.0011$$

$$P(X=3) = \frac{e^{-9} (9)^3}{3!} = 0.0050$$

* For 2-minute interval:

$$\lambda = 4.5 \frac{\text{hits}}{\text{min}} \cdot 2 \text{ min} = 9 \text{ hits}$$

$$* P(X=4) = \frac{e^{-9} (9)^4}{4!} = 0.0337$$

$$* P(X=5) = \frac{e^{-9} (9)^5}{5!} = 0.0607$$

$$* P(X=6) = \frac{e^{-9} (9)^6}{6!} = 0.0911$$

$$* P(X=7) = \frac{e^{-9} (9)^7}{7!} = 0.1171$$

$$* P(X=8) = \frac{e^{-9} (9)^8}{8!} = 0.1318$$

$$* P(X=9) = \frac{e^{-9} (9)^9}{9!} = 0.1318$$

$$P(X=10) = \frac{e^{-9} (9)^{10}}{10!} = 0.1186$$

$$P(X > 10) = 1 - (0.0001 + 0.0011 + 0.0050 + 0.0150 + 0.0337 + 0.0607 + 0.0911 + 0.1171 + 0.1318 + 0.1318 + 0.1186)$$

$$P(X > 10) = 0.294 \text{ or } 29.4\%$$

4. Continuous Probability & Normal Distributions

(a) What proportions of queries take longer than 135 milliseconds to process?

Given: $\mu = 120$ milliseconds

$\sigma = 15$ milliseconds

* Get z-value: $z = \frac{x - \mu}{\sigma}$

$$z = \frac{135 - 120}{15} = 1$$

$$P(X > 135) = P(Z > 1)$$

From the z-table: $P(Z < 1) = 0.8413$

* Since the value from z-table gives area to the left, subtract 1 to get

$$P(Z > 1)$$

$$P(Z > 1) = 1 - 0.8413 = \boxed{0.1587 \text{ or } 15.87\%}$$

(b) Find the probability that a randomly chosen query takes between 100 and 130 milliseconds.

$$z = \frac{100 - 120}{15} = -1.33$$

$$z = \frac{130 - 120}{15} = 0.67$$

$$P(100 < X < 135) = P(-1.33 < Z < 0.67)$$

$$P(Z = 0.67) = 0.7486$$

$$P(Z = -1.33) = 0.0918$$

$$P(-1.33 < Z < 0.67) = 0.7486 - 0.0918$$

$$= 0.6568 \text{ or } 65.68\%$$

(c) Find the 95th percentile of query processing times.

$$P(X = x) = 95$$

* From the z-table, the z-value that gives an area closest to 0.95 is 1.65

$$z = \frac{x - \mu}{\sigma}$$

$$1.65 = \frac{x - 120}{15}$$

$$24.75 = x - 120$$

$$x = 144.75$$